

GraphQL Blog — Configuration and Environment

Project: GraphQL Blog

Technology: Python, FastAPI, Strawberry GraphQL, SQLAlchemy, PostgreSQL

Location: /home/dominic/Documents/graphql-blog

Report #: 8/20

Environment + Docker Compose

```
services:
  db: postgres:15
  app: build: . ports: 8000:8000
  nginx: ...
```

PostgreSQL connection via environment variables.

Configuration & Environment

Environment Files

```
# .env (example)
DATABASE_URL=sqlite:///./data.db
SECRET_KEY=your-secret-key-here
DEBUG=false
CORS_ORIGINS=*
```

Configuration Loading

```
# Pattern used across projects
from pydantic_settings import BaseSettings

class Settings(BaseSettings):
    database_url: str = "sqlite:///./data.db"
    secret_key: str = "change-me"
    debug: bool = False
    cors_origins: str = ""

    model_config = {"env_file": ".env", "env_file_encoding": "utf-8"}
```

Runtime Configuration

Parameter	Source	Default	Description
DATABASE_URL	.env	sqlite:///./data.db	DB connection
SECRET_KEY	.env	(required)	JWT signing
DEBUG	.env	false	Dev mode
CORS_ORIGINS	.env	*	CORS policy

Deployment-Specific Config

Render Web Service (free tier): - CPU: Shared - RAM: 512 MB - Disk: 1 GB - Region: Frankfurt - Auto-deploy: On push to main

Sensitive Values

Variable	Risk	Mitigation
SECRET_KEY	JWT forgery	Use strong random value, rotate periodically
OPENAI_API_KEY	Cost, data exposure	Server-side only, rate-limited
DATABASE_URL	Data access	Never commit to git